

Civics Group	Index Number	Name (use BLOCK LETTERS)
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H2



ST. ANDREW'S JUNIOR COLLEGE
2017 JC2 BT2

H2 BIOLOGY

9744/1

Paper 1: Multiple Choice Mark Scheme

Tuesday

19th September 2017

1 hour

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser (not supplied)
Soft pencil (type B or HB is recommended)

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, civics group and index number on the multiple choice answer sheet in the spaces provided.

There are **30** questions in this paper. Answer all questions. For each question, there are four possible answers, A, B, C and D.

Choose the one you consider correct and record your choice in soft pencil on the separate multiple choice answer sheet.

INFORMATION TO CANDIDATES

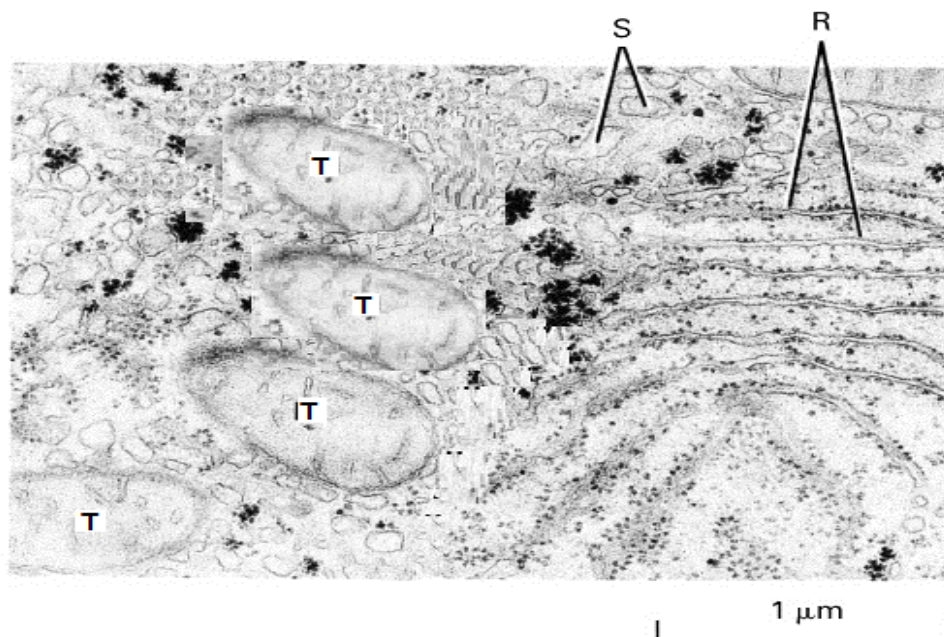
Each correct answer will score one mark. A mark will not be deducted for wrong answer. Any rough working should be done in this booklet.

At the end of the examination, submit both question paper and multiple choice answer sheet.

This document consists of **16** printed pages.

[Turn over

1 The figure below shows an electron micrograph of an eukaryotic cell.



Which of the following option correctly matches the structures **R**, **S** and **T** to their respective functions?

	R	S	T
A	Involved in proteins glycosylation	Site of lipid synthesis	To convert light energy to chemical energy
B	Site of protein synthesis	Site of detoxification reaction	Supplying cellular energy
C	Site of detoxification reaction	Involved in protein glycosylation	Remove worn out organelles
D	Site of protein synthesis	Contains proteins to be secreted	Storage of starch

2 Which comparative statement(s) concerning biological molecules is/are correct?

- 1 A collagen molecule is a fibrous protein that contains many amino acids with hydrophobic R-groups whereas a haemoglobin molecule is a globular protein with no amino acids with hydrophobic R-groups.
- 2 Sucrose hydrolysis results in glycosidic bond breakage and the production of equal proportions of fructose and α -glucose molecules, whereas cellulose hydrolysis results in only β -glucose molecules.
- 3 The glycosidic bonds of glycogen are formed between two α -glucose molecules, whereas in amylopectin, the bonds are formed between an α -glucose molecule and a β -glucose molecule.

- A** 2 only
- B** 3 only
- C** 1 and 2
- D** 1 and 3

3 The statements below are about bonds found in biological molecules.

- 1 They are formed by condensation.
- 2 Oxygen is part of the bond.
- 3 ATP is hydrolysed to form the bonds.
- 4 They can be broken at 100°C.

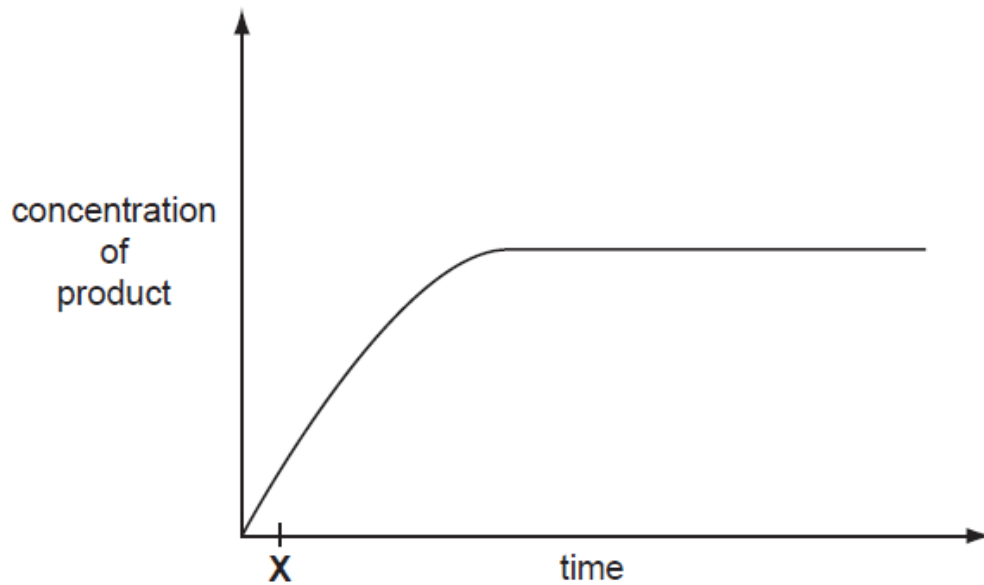
Which statements are correct for the bonds in the primary structure of proteins?

- A** 1 and 2 only
- B** 1 and 3 only
- C** 3 and 4 only
- D** 1, 2 and 4 only

4 What supports the view that a membrane protein is involved in active transport?

- A** It allows movement of molecules across a membrane if concentration differences exist.
- B** It can only function if mitochondria are supplied with sufficient oxygen.
- C** It has a tertiary structure with a binding site with a specific shape.
- D** It is found in the cell surface membranes and the mitochondrial membranes.

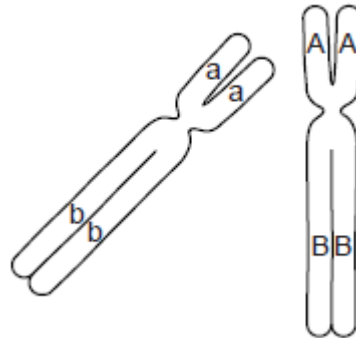
- 5 The graph shows the course of an enzyme-catalysed reaction at 30 °C.



What is true at time X?

- A** Most enzyme molecules will have free active sites.
- B** The number of available substrate molecules is high.
- C** The number of enzyme-substrate complexes is low.
- D** The rate remains the same if more enzyme is added.

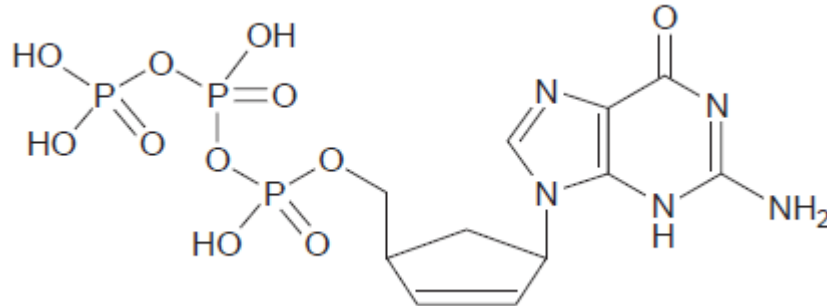
- 6 The diagram shows two homologous chromosomes in early prophase I of meiosis in an animal cell. Two genes, **A/a** and **B/b**, whose loci occur on the homologous chromosomes are also shown.



Which row of diagrams is a possible representation of these chromosomes as they progress from anaphase I to prophase II? **ANS: D**

	anaphase I	prophase II
A		
B		
C		
D		

- 7 The diagram shows the molecular structure of a chemical that can inhibit the activity of reverse transcriptase. It is an analogue of a naturally occurring nucleic acid monomer.



Which option is correct?

	Analogue	Naturally occurring monomer
A	Acts as a competitive inhibitor	Is an activated DNA nucleotide
B	Acts as a non-competitive inhibitor	Is an activated RNA nucleotide
C	Acts as a competitive inhibitor	Is an activated RNA nucleotide
D	Acts as a non-competitive inhibitor	Is an activated DNA nucleotide

- 8 In dogs, a gene on chromosome 27 is responsible for the curliness of the dog's hair. One form of this gene produces an enzyme with arginine at residues 151, but a mutant allele of the gene produces an enzyme which has cysteine at this point.

This latter form causes kinks in the keratin so that the coat is curlier. In heterozygotes, both alleles are co-dominant so an intermediate 'wavy' coat can be observed in the phenotype.

In this context, what is meant by gene mutation?

- A** change in the gene locus
- B** chromosome 27 inversion
- C** production of a new protein
- D** structural change in DNA

- 9 Individuals with the rare Bombay phenotype (i.e. homozygous recessive, hh) do not produce H-antigen. As a result, they cannot make A-antigen or B-antigen on their red blood cells, regardless of what alleles they may have at the gene locus which determines the ABO blood group. This is because A-antigens and B-antigens are synthesised from H-antigens. Individuals with blood group O do not produce A-antigens or B-antigens but produce H-antigens on their red blood cells.

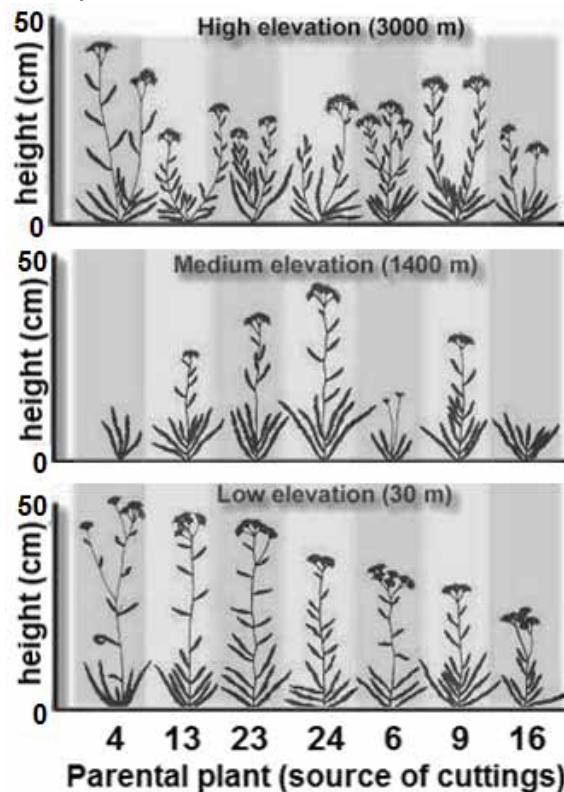
Alan and his family had the following phenotypes in their blood groups:

Family member	Blood type
Alan's father	O
Alan's mother	Bombay phenotype
Alan	A
Alan's wife	Bombay phenotype
Alan's wife's father	O
Alan's wife's mother	O

If Alan and his wife had a child, what is the probability that the child will have Bombay phenotype?

- A 0
- B $\frac{1}{2}$
- C $\frac{1}{4}$
- D $\frac{3}{4}$

- 10 Seven plants of the same species were selected from a field. Three plant cuttings were obtained from each parent plant, each of which were planted at various elevations on a mountain terrain. They were allowed to grow for 3 months. Their heights and flower development were shown in the table below.



Which of the following statements can be deduced from the results of the experiment?

- A The height of the plants is genetically determined.
- B The development of flowers is affected by environmental factors.
- C Genetic variation exists in the population of plants in the field due to genetic drift.
- D Plants that did not develop flowers at medium elevation have mutations that inhibit flower development.

11 Which of the following is not the consequence of natural selection?

- A The field mustard plant survived a summer drought in southern California because some individuals contained alleles that made them flower earlier. Plants with flowers wilt more easily than plants without flowers. Now, almost all the field mustard plants in California flower in spring.
- B In areas with fewer predators of herbivorous insects, plants which produce higher concentrations of alkaloids (which are toxic to insects) dominated the landscape. Most of the herbivorous insects in these areas are found to be able to accumulate alkaloids in their bodies without affecting their metabolism.
- C Endemic to New Zealand, the kakapo (a large flightless bird) had no natural predators before the humans arrived. They have evolved to have very few offspring throughout their entire lifespan. This phenomenon is also common for other island species which do not have natural predators in their respective habitats.
- D Maple probably has the most variation in bark of any tree species. Japan experiences tornadoes which destroy large trees like the maple. Over the last few decades, it was observed that only the Japanese maple with dark-coloured bark remained.

12 Neanderthals evolved from *Homo erectus* via the intermediate *H. heidelbergensis*; this *H. erectus* lineage left Africa about 600kya (kya = 1000 years ago), the descendants made their way to Europe and the earliest *H. heidelbergensis* fossils appeared 500kya. The *H. erectus* line continues in Europe after the *H. heidelbergensis* speciation, with the later fossils being found in France (450kya), Hungary (350kya), Indonesia (250kya) and China (210kya).

In contrast, *H. sapiens* evolved in east Africa from *H. erectus* through the intermediate form of *H. rhodesiensis*. *H. rhodesiensis* fossils range from 600–200kya.

Based solely on the information given above, which of the following statements is least likely to be true?

- A *H. erectus* and *H. rhodesiensis* are unable to interbreed to produce viable and fertile offspring.
- B Geographical isolation resulted in the disruption of gene flow between Neanderthals and *H. sapiens*.
- C The divergence of *H. rhodesiensis* from *H. erectus*, and *H. sapiens* from *H. rhodesiensis* is sympatric speciation.
- D The fossils of *H. erectus* found in France share more similarities with those found in China than with the earliest fossils of *H. heidelbergensis*.

- 13** The plica semilunaris is a small fold of tissue on the inside corner of the eye. It is the vestigial remnant of the nictitating membrane, an organ that is fully functional in some other species of mammals. For example, in diving animals like beavers and manatees, the nictitating membrane is transparent and moves across the eye to protect it while under water.



Which of the following statements least explains the presence and structure of plica semilunaris in humans?

- A** Early ancestors of humans were not divers.
- B** Any presence of nictitating membrane in non-diving mammals posed a selective disadvantage for individuals who had it.
- C** Mutations occurred to reduce the size of nictitating membrane in humans to its present-day vestigial structure as there was no use for it.
- D** The genes involved in producing the plica semilunaris were inherited from a common ancestor shared by humans, beavers and manatees.

Explanation:

Early ancestors of humans were not divers. That's why our nictitating membrane is now vestigial, rather than a fully functional one.

The fact that we inherited it despite not being divers explains why we still have this vestigial remnant, instead of not having it at all.

- 14** Which of the following correctly shows the effects of climate change on coral reefs and associated ecosystems?

	Average number of zooxanthellae in each polyp	Mass of basal plate of hard corals	Diversity of catch from nearby fisheries
A	Decreased	Decreased	Decreased
B	Decreased	Unaffected	Increased
C	Increased	Decreased	Decreased
D	Increased	Unaffected	Increased

- 15** Which of the following is true of the pathogenicity of the dengue virus?

- A** Infected cells release interferons which cause dengue fever.
- B** Infected cells release toxins which kill cytotoxic T cells and red blood cells.
- C** Dengue virus infects and kills macrophages, causing dengue shock syndrome.
- D** Antibodies specifically recognise and neutralise the dengue viruses circulating in the circulatory and lymphatic systems.

- 16** The enzyme DNase is added to samples taken from the same chromosomal region from three different cell types. After 20 minutes, the remaining DNA samples are weighed and the following results are obtained.

Cell type	Mass of DNA before adding DNase/ μg	Mass of DNA after adding DNase/ μg
P	5.0	5.0
Q	5.0	3.4
R	5.0	4.1

Which of the following statements correctly suggests about the genes and their expressions in the three different cell types?

- A** Some regions of the cell type **P** chromosomes contain DNA coding for rRNA.
- B** The mass of DNA in cell type **P** did not decrease because the histone tails are acetylated.
- C** There are more phosphodiester bonds in the DNA of cell type **Q** than those in cell types **P** and **R**.
- D** The genes found in the same chromosomal region in the three cell types are expressed the most in cell type **Q**.

17 The following statements are about eukaryotic control elements:

- 1 Attachment of RNA polymerase II at the TATA box is achieved with the help of a series of specific transcription factors.
- 2 The DNA binding site on general transcription factors and specific transcription factors is different in DNA sequence.
- 3 When the histones found in part of a chromosome are acetylated, the control elements of a gene are easily accessed.
- 4 Repressors bind to regions of DNA to repress transcription.

Which of the above statement(s) is/are true?

- A** 3 only
- B** 1 and 2
- C** 3 and 4
- D** 2, 3 and 4

- 18** Transcription in eukaryotic cells results in the formation of pre-mRNA, which is made up of exons and introns.

Which of the following statements correctly describes what happens during the formation of mature mRNA from the pre-mRNA?

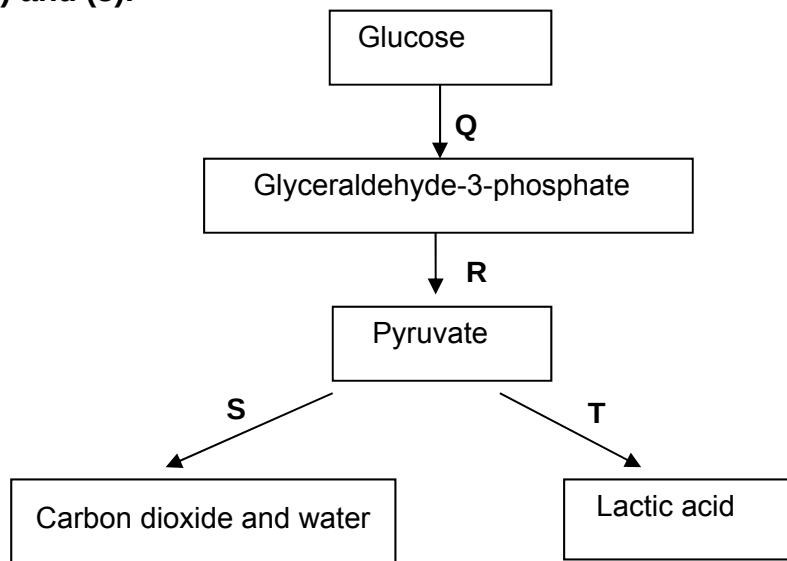
- A** The 5' of the intron is cut, and joined to the branch-point sequence, followed by the cutting of the 3' end to form the lariat loop.
- B** RNA splicing occurs, where all introns are recognised as they share highly similar sequences and are excised.
- C** RNA splicing occurs, where all the introns are excised and some of the exons joined together so that they can be transcribed.
- D** The addition of the 5' cap and the 3' poly-A tail occurs, followed by RNA splicing.

- 19** Regulation of gene expression occurs in both prokaryotes and eukaryotes.

Which of the following process(es) is/are involved in both prokaryotes and eukaryotes?

- 1 extension of telomeres by telomerase
 - 2 down-regulation by repressor molecules
 - 3 post-translational modification
 - 4 alternative splicing
- A** 2 only
 - B** 1 and 3 only
 - C** 2 and 4 only
 - D** 2, 3 and 4 only

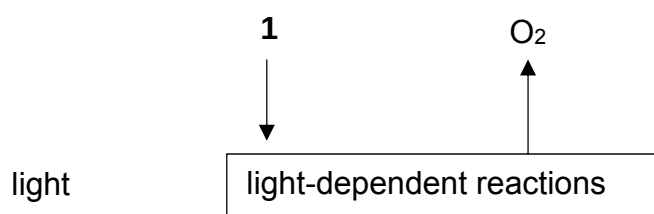
20 With reference to the diagram below, relate processes **P, Q, R, S, T** to statements **(1), (2) and (3)**.

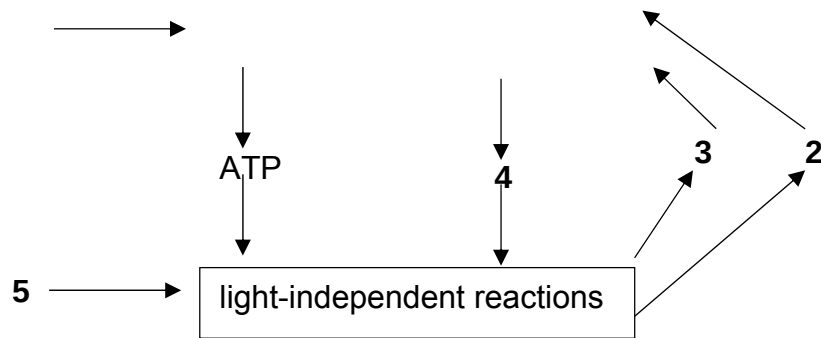


- (1) NAD is regenerated without the use of the electron transport system
- (2) ATP is synthesised via substrate level phosphorylation
- (3) It can take place under anaerobic conditions.

	(1)	(2)	(3)
A	T only	R only	Q,R,T only
B	T only	R,S only	Q,R,T only
C	S,T only	R only	Q,R,S,T
D	S,T only	R,S only	Q,R,S,T

21 The diagram summarises the process of photosynthesis.

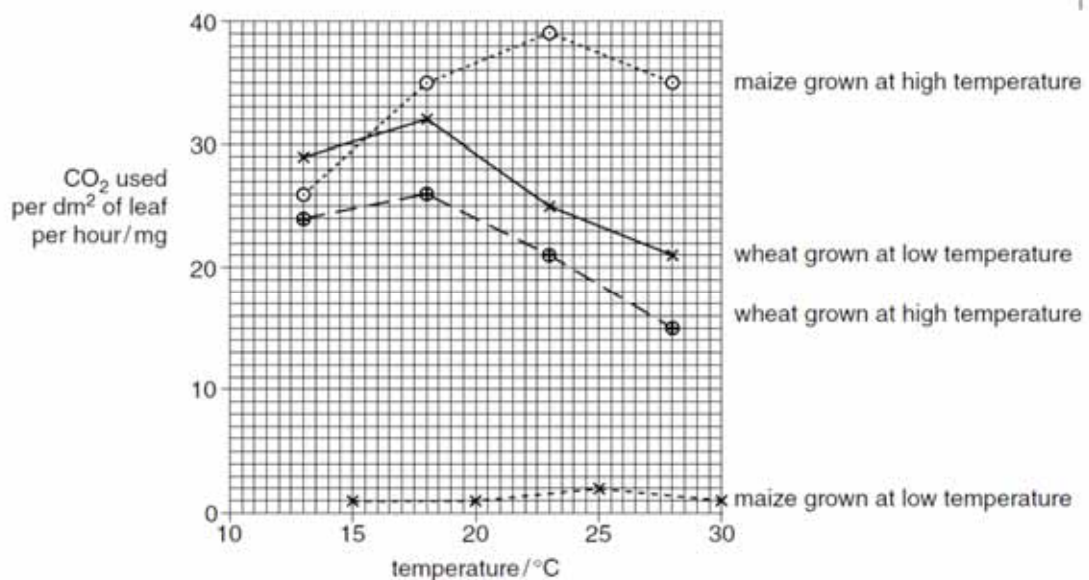




Which row identifies the reactants **1,2,3, 4** and **5**?

	1	2	3	4	5
A	Carbon dioxide	ADP + phosphate	reduced NAD	NAD	water
B	Carbon dioxide	reduced NAD	ADP + phosphate	NADP	water
C	water	NAD	reduced NAD	ADP + phosphate	Carbon dioxide
D	water	NADP	ADP + phosphate	reduced NADP	Carbon dioxide

22 Young maize and wheat plants were grown to maturity at high and low temperatures. The rate of photosynthesis in each of these mature plants was measured at different temperatures. The rate of photosynthesis was measured as the amount of CO_2 used per dm^3 of leaf per hour. The results are shown in the graph below



What information can be concluded from the graph above?

- 1 For plants grown at high temperature, the rate of photosynthesis is optimum at 25°C in maize and 18°C in wheat.
- 2 For plants grown at high temperature, maize had a greater increase in rate of photosynthesis compared to wheat until optimum temperature was reached.
- 3 The rate of photosynthesis was affected more significantly in maize plants than in wheat plants when grown at low temperatures.
- 4 Low temperatures slowed down the formation of membranes in maize plants but not in wheat plants which caused a decrease in lamellae formation.

- A** 2 and 3 only
B 1 and 4 only
C 1, 2 and 3 only
D All of the above

23 Which of the following pairs of statements is **true** of transduction and conjugation?

	Transduction	Conjugation
A	Bacterial DNA is transferred from donor cell to recipient cell	Bacterial DNA is transferred from donor cell to recipient cell
B	Only host DNA adjacent to prophage is transferred from donor cell to recipient cell in specialised transduction	F plasmid is exchanged between donor cell and recipient cell
C	Lambda lysogenic phage is involved in generalised transduction	T4 lytic phage is involved
D	Viral DNA is replicated via rolling-circle mechanism in the donor cell	DNA on F plasmid is replicated via rolling-circle mechanism in the donor cell

24 All of the following statements about viruses are true **except**:

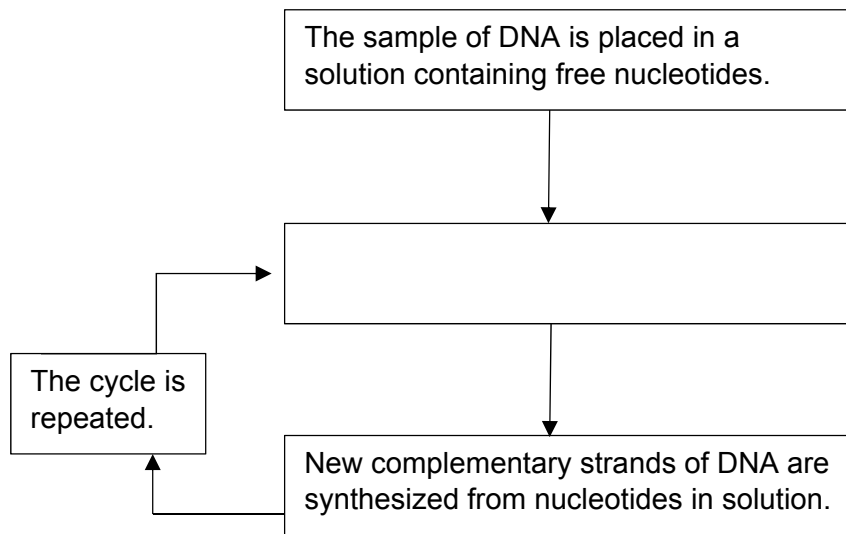
- A** The genome of HIV is more likely to mutate than the genome of bacteriophages
- B** Before entering a host cell, specific proteins of viruses bind to receptors on specific host cells.
- C** All viruses produce RNA as an intermediate molecule during the production of new viruses.
- D** HIV and influenza viruses produce DNA as an intermediate molecule during the production of new viruses.

25 Arrange the following statements on signal transduction pathway for insulin in order.

- 1 Auto-crossphosphorylation
- 2 Increase in uptake of glucose through facilitated diffusion
- 3 Relay proteins bind to specific activated tyrosine residues
- 4 Activated relay proteins activate their respective transduction pathways
- 5 Insulin binds to receptor tyrosine kinase (RTK) at the receptor site
- 6 Vesicles containing glucose transporters move to and fuse with the plasma membrane
- 7 Changes in the 3D conformation activates the tyrosine kinase domain of receptor

- A** 5, 1, 7, 3, 4, 6, 2
B 5, 7, 1, 3, 4, 6, 2
C 2, 5, 1, 7, 3, 4, 6
D 2, 5, 1, 7, 4, 3, 6

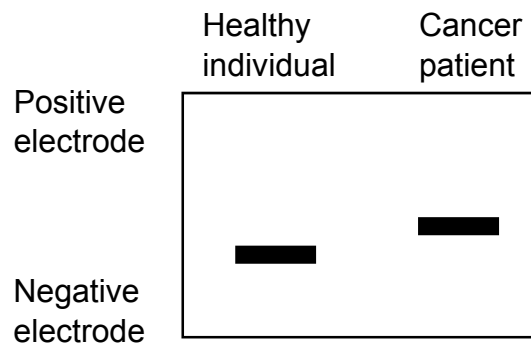
26 The polymerase chain reaction is summarised in the flowchart below.



Which statement completes the flow chart?

- A** Complementary strands of DNA are separated.
- B** Free nucleotides join on the end of DNA strands.
- C** Small sections of DNA are formed.
- D** Strands of DNA bind to RNA primers.

27 A gene involved in the development of cancer was studied using the technique of gel electrophoresis.



Which of the following can possibly be concluded from the results of this study?

- A** Mutation of gene resulting in a hyperactive protein
- B** Additional DNA nucleotides are inserted within the mutant allele
- C** Amplification of gene related to cancer
- D** Over-expression of the gene related to cancer

28 Some of the features of different types of stem cells are listed.

- 1 They are able to develop into all cell types of the body to form a whole organism
- 2 They can develop into a wide range of different types of cell
- 3 They have active telomerase enzyme
- 4 They can only develop into a limited range of cell types

Which of the following will be shown by embryonic stem cells?

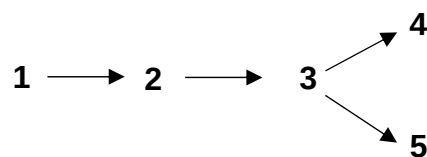
- A** 1 and 2
- B** 1 and 3
- C** 2 and 3
- D** 3 and 4

29 Most cases of cervical cancer is caused by infection with Human Papilloma Virus (HPV). Vaccines consisting of HPV antigens are available for prevention of cervical cancer.

Which of the following statements is true?

- A** The HPV vaccine offers passive artificial immunity.
- B** The HPV vaccine offers active natural immunity.
- C** HPV vaccine stimulates the production of specific antibodies by T cells
- D** HPV vaccine will be ineffective against the influenza virus due to different 3D conformation from influenza viral antigens

30 1-5 represents different components of the immune system. Arrows may represent processes such as activation of other cells and differentiation of cells. Which of the following could be a possible representation of 1-5?



	1	2	3	4	5
A	Antigen-Presenting Cell	T cytotoxic cell	Antibodies	B cell	Memory B cell
B	Antigen-Presenting Cell	B cell	T helper cell	T cytotoxic cell	Memory T cell
C	Antigen-Presenting Cell	T helper cell	B cell	Plasma cell	Antibodies
D	Antigen-Presenting Cell	T helper cell	B cell	Plasma cell	Memory B cell

SAJC H2 PRELIM 2017 PAPER 1
ANSWER SCHEME

1	B	11	D	21	D
2	A	12	D	22	A
3	B	13	C	23	A
4	B	14	A	24	D
5	B	15	A	25	B
6	D	16	D	26	A
7	A	17	C	27	A
8	D	18	A	28	C
9	B	19	A	29	D
10	B	20	B	30	D